

defn. of conditional, $P(a) \rightarrow A$ is False. By the defn. of universal quantifier, $\forall x(P(x) \rightarrow A)$ is False.

$$\therefore \exists x(P(x) \rightarrow A) \text{ is False} \Rightarrow \forall x(P(x) \rightarrow A) \text{ is False}$$

$\therefore$ They are logically equivalent $\square$

$$b) \exists x(P(x) \rightarrow A) \equiv \forall x P(x) \rightarrow A$$

Let, $\exists x(P(x) \rightarrow A)$ be False. By defn. of existential quantifier, for all elements a in the domain, $P(a) \rightarrow A$ is False. By defn. of conditional, $P(a)$ is true for all elements a in the domain and A is False. By defn. of universal quantifier, $\forall x P(x)$ is ~~True~~ True. By the defn. of conditional, $\forall x P(x) \rightarrow A$ is False.

$$\therefore \exists x(P(x) \rightarrow A) \text{ is False} \Rightarrow \forall x P(x) \rightarrow A \text{ is False}$$

Let, $\forall x P(x) \rightarrow A$ be False. By the defn. of conditional $\forall x P(x)$ is True and A is False. By the defn. of universal quantifier, for all elements a in the domain, $P(a)$ is True. By the defn. of conditional, $P(a) \rightarrow A$ is False for all elements a in the domain. By the defn. of existential quantifier, $\exists x(P(x) \rightarrow A)$ is False.

$$\therefore \forall x P(x) \rightarrow A \text{ is False} \Rightarrow \exists x(P(x) \rightarrow A) \text{ is False}$$

$\therefore$ They are logically equivalent $\square$

52) Show that $\forall x P(x) \vee \forall x Q(x)$ and $\forall x(P(x) \vee Q(x))$ are not logically equivalent.

Soln: Let, $\forall x P(x) \vee \forall x Q(x)$ be False. By defn. of OR, $\forall x P(x)$ is False and $\forall x Q(x)$ is False. By defn. of universal quantifier, ~~there exists~~ there exists at least one element a_1 in the domain s.t. $P(a_1)$ is False. By defn. of universal quantifier, there exists at least one element a_2 in the domain s.t. $Q(a_2)$ is False. By defn. of OR, $P(a_1) \vee Q(a_1)$ has the same truth value as $Q(a_1)$. Similarly, by defn. of OR,